

6G Weekly Survey — Jun 01 - Jun 07, 2026

18 items across 6 topics, ranked by importance.

6G Physical Layer

LO-Free Joint Communication and Sensing via Inter-Antenna Cross-Correlation and Graph-Based Spatial Phase Inference

Hasan Atalay Günel, Mohaned Chraïti, Ali Görçin · arXiv eess.SP · 2026-06-01

★★★★■ Addresses a critical implementation barrier (LO cost and power) for mmWave/sub-THz ISAC systems with a novel graph-theoretic framework, highly relevant for practical 6G hardware design.

The paper proposes an LO-free joint communication and sensing receiver for mmWave/sub-THz that uses pairwise inter-antenna correlations to recover relative phase relationships across the array. Symbol design encodes both data and DoA information into correlation phases, which are recovered via graph-based inference exploiting cycle consistency. Numerical results show improved BER and DoA accuracy with increased graph connectivity, approaching derived Cramér-Rao bounds.

Contribution: First demonstration of coherent array processing for JCAS without local oscillators by reformulating phase recovery as inference over a correlation graph with cycle-consistent redundancy.

Relevance to your work: Directly relevant: tackles hardware complexity reduction for ISAC at sub-THz frequencies while enabling DoA-based sensing through learned phase relationships, connecting to your interests in ISAC and sub-THz physical layer design.

Electromagnetic Digital Twin-Enabled Closed-Loop Beam Management in ISAC Systems

Yubin Luo, Takumi Takahashi, Li Yu, Zhaohui Yang, et al. · arXiv eess.SP · 2026-05-31

★★★★■ Addresses critical gap between DT construction and control with practical array calibration, from credible ISAC research groups, though experimental validation on real hardware would strengthen impact.

This paper proposes a closed-loop digital twin network (DTN) framework for ISAC systems that jointly reconstructs electromagnetic scatterer properties and calibrates array geometry mismatches using Bayesian message-passing inference. The DT guides beam codebook preselection and channel prediction for low-latency beamforming control. Simulations show improved robustness to array calibration errors compared to existing open-loop DT approaches.

Contribution: First closed-loop DTN framework integrating EM-based environment reconstruction with joint array calibration and DT-driven beam management, addressing the practical challenge of array geometry mismatch in ISAC-enabled digital twins.

Relevance to your work: Directly relevant to your ISAC and beam management interests, demonstrating how sensing-enabled digital twins can reduce CSI overhead and enable proactive beamforming—core elements of AI-native 6G air interfaces.

Radar-Assisted Beam Management Framework for mmWave NTNs: Overhead Reduction and Physical Layer Security Application

Bora Bozkurt, Cevdet Tosun, Mehmet Nuri Akinci, Ali Gorcin, et al. · arXiv eess.SP · 2026-05-29

★★★★■ Solid contribution addressing practical NTN beam management overhead, but limited to idealized analysis and simulation without real-world validation or large-scale evaluation.

This paper proposes using radar-derived spatial information (AoD, distance) to reduce beam selection overhead in mmWave non-terrestrial networks. A probabilistic analysis quantifies worst-case overhead under Gaussian error, and the framework is extended to physical-layer security by detecting passive eavesdroppers. Simulations show >135 dB suppression of unintended user power and 2 dB gain for legitimate users.

Contribution: First radar-assisted beam management framework for NTN with theoretical overhead analysis and demonstrated PLS application via passive target detection.

Relevance to your work: Directly relevant: combines NTN beam management with integrated sensing (radar), addressing overhead reduction critical for LEO/satellite systems and extending to security applications aligned with 6G ISAC trends.

ISAC (sensing + communication)

Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems

Lu Wang, Mohamadreza Delbari, Gui Zhou, Luis F. Abanto-Leon, et al. · arXiv eess.SP · 2026-06-02

★★★★■ Addresses a critical but overlooked practical deployment issue for RHS-ISAC with credible performance bounds and optimization theory, highly relevant as RHS matures toward 6G trials.

This paper addresses hardware faults (uncontrollable amplitude elements) in reconfigurable holographic surfaces (RHS) deployed for integrated sensing and communication systems. The authors derive the misspecified Cramér-Rao bound for sensing and SINR for communication to quantify fault-induced performance loss, then formulate a joint optimization problem minimizing MCRB under SINR and power constraints. A block coordinate descent algorithm incorporating majorization-minimization achieves 13.7% average gain over fault-unaware baselines.

Contribution: First work to rigorously model and compensate for faulty RHS elements in ISAC systems using misspecified CRB theory and a tractable convex-relaxation-based optimization framework.

Relevance to your work: Directly relevant to your ISAC research—demonstrates how physical-layer hardware imperfections impact joint radar-communication performance and provides optimization techniques applicable to robust beam management and surface configuration design.

SNF-PRP: A Covert Integrating Sensing and Communications Framework

Dhruvil Bhatt, Vidushi Kumar · arXiv eess.SP · 2026-06-02

★★★★■ Addresses a legitimate security gap in ISAC with rigorous analysis and 5G-compliant validation, though covertness against sophisticated adversaries and practical deployment challenges remain open questions.

This paper proposes sub-noise-floor pseudo-random probing (SNF-PRP) for OFDM-based ISAC systems that hides sensing signals below the noise floor to prevent passive wardens from detecting sensing activity. The method provides ϵ -covert guarantees via KL divergence bounds, exploits multi-subcarrier spreading gain, and derives closed-form expressions for integration length needed to achieve target Cramér-Rao bounds. 5G NR simulations demonstrate sub-0.5m range/velocity accuracy at -12 to -15 dB probing power while maintaining covertness $5.8\times$ below detection thresholds.

Contribution: First framework to achieve provably covert sensing in OFDM ISAC by operating below the noise floor with formal KL-divergence covertness guarantees and closed-form design rules linking integration time to estimation accuracy.

Relevance to your work: Directly relevant to your ISAC research by introducing physical-layer security considerations and offering design principles for sub-noise-floor operation that could inform future JCAS waveform design and standard discussions.

On Secure EKF-enhanced UAV-ISAC Systems

Hongjiang Lei, Heng Jin, Ki-Hong Park, Jia Ye, et al. · arXiv cs.IT · 2026-06-02

★★★★■ Solid incremental contribution applying standard optimization and filtering techniques to UAV-ISAC security, well-motivated but using conventional methods without fundamental breakthroughs.

This paper addresses secure UAV-assisted ISAC by jointly optimizing transmit beamforming and UAV trajectory to maximize secrecy rate while tracking legitimate users and jamming eavesdroppers. An extended Kalman filter (EKF) is integrated to predict and track user trajectories under practical constraints including sensing beamwidth, transmit power, and propulsion energy. The non-convex optimization is solved via block

coordinate descent and successive convex approximation with EKF integration.

Contribution: First work to combine EKF-based trajectory prediction with joint beamforming and UAV trajectory optimization for secure ISAC, explicitly handling sensing beamwidth constraints.

Relevance to your work: Directly relevant to ISAC research, particularly the interplay between sensing (tracking), communication, and physical-layer security in mobile scenarios with resource constraints.

Instantaneous Risk Minimization for Secure Integrated Sensing and Communication

Chao Ge, Na Zhao, Yuan Shen · arXiv eess.SP · 2026-06-02

★★★★ Solid work addressing security in ISAC with a practical robust optimization framework, though security aspects are less central to current ISAC standardization discussions than fundamental sensing-communication trade-offs.

The paper addresses physical-layer security in ISAC systems by proposing a robust beamforming framework that minimizes instantaneous information leakage risk rather than maximizing ergodic secrecy rate. A hierarchical iterative algorithm is developed where the outer loop refines uncertainty regions based on sensing performance and the inner loop optimizes beamforming under these constraints. The method provides per-transmission security guarantees while jointly managing eavesdropper SINR suppression, sensing accuracy, and legitimate user QoS.

Contribution: Novel worst-case instantaneous risk minimization formulation for secure ISAC that provides per-transmission security guarantees, departing from conventional ergodic secrecy rate maximization approaches.

Relevance to your work: Directly relevant to your ISAC work as it addresses the joint design problem with a novel security-focused perspective, though the robust beamforming methodology is more incremental than your likely focus on fundamental ISAC architectures and sensing modalities.

Integrated Sensing and Covert Communication In Low-Altitude Networks: A Smart Radio Environment Perspective

Jianyu Wei, Haichao Wang, Laixian Peng, Jiangchun Gu, et al. · arXiv eess.SP · 2026-06-01

★★★★ Solid application-driven work combining emerging ISAC concepts with covert communication and SRE, though the security focus is narrower than mainstream ISAC physical-layer research and demonstration is simulation-only without real channel validation.

This paper addresses covert communication in low-altitude networks by combining integrated sensing and communication (ISAC) with smart radio environment (SRE) techniques to hide transmission behavior while maintaining sensing functionality. The authors propose a reinforcement learning framework that jointly optimizes flight trajectory, jamming power, movable antennas, bandwidth allocation, and beamforming to maximize covert throughput in dynamic urban environments. Simulations demonstrate performance gains over baseline approaches in this integrated sensing and covert communication (ISACC) scenario.

Contribution: Novel integration of smart radio environment technologies (movable antennas, intelligent surfaces) with ISAC for covert communication in low-altitude networks, including a joint optimization framework for multiple physical-layer parameters.

Relevance to your work: Directly relevant to your ISAC research interests, particularly in understanding how intelligent environment control and multi-objective optimization (sensing + security) extend the ISAC design space for low-altitude and UAV scenarios.

NTN / satellite

Mapping the Storm: Geospatial Impacts of Severe Weather on LEO Network Performance

Sina Ehsani, Bhanu Pallakonda, Pragyana K. Mishra · arXiv cs.NI · 2026-06-01

★★★★■ Provides unique large-scale empirical dataset and geospatial methodology for NTN reliability analysis, directly relevant to 3GPP NTN standardization and LEO service-level design as commercial deployments scale.

This paper analyzes 870,000+ terminal-hours of Starlink telemetry across 1,292 ground terminals in the contiguous US, correlating minute-level KPIs (latency, drop rate, signal quality) with high-resolution weather observations. Using spatial join techniques and temporal alignment with classified weather events, the authors find that severe weather (heavy rain/snow thunderstorms) causes substantial degradation in >55% of affected terminals, with outages lasting minutes to hours. The work provides the first continental-scale empirical characterization of weather-induced LEO network impairment and proposes a framework for weather-aware provisioning and geospatial forecasting.

Contribution: First large-scale empirical study linking fine-grained spatio-temporal weather data to LEO satellite Internet performance degradation using real Starlink telemetry at minute-level granularity across a continental footprint.

Relevance to your work: Directly relevant to your NTN research focus: establishes empirical foundations for weather-aware beam management, predictive handover strategies, and geospatial channel modeling in LEO direct-to-device scenarios.

Toward Agile and Cooperative LEO Satellite Beam-Hopping Networks: Paradigms, Challenges, and Opportunities

Xinyi Huang, Bodong Shang, Meixia Tao · arXiv eess.SP · 2026-05-30

★★★★■ Addresses critical NTN capacity allocation challenges with cooperative transmission mechanisms highly relevant to 3GPP NTN evolution and commercial LEO constellations (Starlink, OneWeb).

This article presents a comprehensive framework for LEO satellite beam-hopping (BH) networks that dynamically allocate capacity to address spatio-temporal traffic non-uniformity. The authors detail BH paradigms including beam pattern design, on-demand scheduling, and cooperative multi-satellite transmission for interference management. A system-level analysis with interference modeling and a cooperative transmission case study demonstrates performance improvements for adaptive coverage, IoT, and delay-sensitive services.

Contribution: Provides the first comprehensive architectural framework integrating cooperative multi-satellite beam-hopping with detailed interference models and performance analysis for LEO networks.

Relevance to your work: Directly relevant to your NTN research focus, particularly for AI-driven beam management and dynamic resource allocation in LEO networks with potential ML applications in scheduling and interference coordination.

Satellite NOMA for Direct-to-Cell Communications: Fundamentals, Protocols, and Opportunities

Xiangyu Li, Bodong Shang, Junchao Ma, Yuzheng Ren, et al. · arXiv eess.SP · 2026-06-01

★★★★■ Solid incremental work combining two active areas (DTC and NOMA), but NOMA's role in 6G remains uncertain given competing technologies like grant-free access and its limited 5G adoption.

This paper explores integrating non-orthogonal multiple access (NOMA) with direct-to-cell satellite communications to address spectrum efficiency and scalability challenges. The authors present architectural fundamentals, protocols for cooperative and multi-tier satellite networks, and conduct comparative analysis between NOMA and orthogonal multiple access schemes. A case study demonstrates performance benefits of NOMA for DTC scenarios.

Contribution: First comprehensive treatment of NOMA protocols tailored to emerging direct-to-cell satellite architectures, including cooperative and multi-tier network configurations.

Relevance to your work: Directly relevant to your NTN research interests, particularly for understanding multiple access strategies in LEO direct-to-cell systems and spectrum-efficient satellite connectivity solutions.

Rethinking NB-IoT Downlink Synchronization for LEO-NTN: A Novel Overhead Reduction Method and Measurement-Based Evaluation

Ömer Lütfü Karakelle, Erhan Karakoca, Bengü Bilgiç Keskin, ■brahim Hökelek, et al. · arXiv eess.SP · 2026-05-30

★★★★■ Solid incremental work addressing a known LEO-NTN deployment bottleneck with a pragmatic, standards-compatible solution validated by measurements.

The paper addresses initial downlink synchronization failures in NB-IoT over LEO satellites caused by large time-varying Doppler shifts exceeding standard UE correction capability. The authors propose a standard-compliant search-space optimization method that reduces acquisition overhead while maintaining synchronization reliability. Real-time LEO measurements validate the approach under realistic conditions.

Contribution: A practical search-space optimization for NB-IoT downlink synchronization that reduces overhead under large LEO Doppler without requiring non-standard UE modifications.

Relevance to your work: Directly relevant to your NTN research interests, particularly LEO direct-to-cell connectivity and the practical challenges of adapting legacy IoT protocols to non-terrestrial deployment scenarios.

SweetFruit: A Two-Stage Mobile Sensing System for Real-Time Fruit Sugar Estimation

Mark Cardamis, Yanxiang Wang, Chun Tung Chou, Wen Hu · arXiv eess.SP · 2026-05-31

★●●●■ Agricultural sensing application with no relevance to wireless communication, PHY layer design, or 6G network research.

SweetFruit is a mobile sensing system for non-contact fruit sugar estimation using two stages: a 3D point-cloud classifier (SF-PointNet) for coarse high/low sugar screening via ToF camera, followed by NIR spectrometry regression (SF-Net) for precise Brix prediction. Tested on apples and strawberries, the system achieves >90% classification accuracy in Stage 1 and 0.57 Brix RMSE in Stage 2 (22% improvement over NIR-only). The approach demonstrates real-time embedded execution with off-the-shelf low-cost sensors for agricultural quality control.

Contribution: First two-stage mobile system combining ToF depth sensing for prescreening with NIR spectrometry for sugar regression, achieving better accuracy than single-modality approaches with practical embedded deployment.

Relevance to your work: tangential

Semantic communication

Toward Reliable Semantic Communication: Beyond Average Performance

Boyuan Li, Mingze Gong, Shuoyao Wang, Suzhi Bi, et al. · arXiv cs.IT · 2026-05-31

★★★★■ Addresses a fundamental gap in semantic communication research that must be resolved before practical deployment, comes from a credible research group, and frames the reliability problem in a way that could influence future protocol design.

This article argues that semantic communication systems are over-optimized for average-case performance while neglecting reliability under adverse channel conditions and transmission uncertainty. The authors review three existing reliability approaches (channel-aware adaptation, robust codec design, HARQ retransmission) and identify their complementary strengths and limitations. They propose two research directions: robust adaptive semantic communication under imperfect CSI and joint source-channel-check coding with adaptive retransmission for sample-level reliability guarantees.

Contribution: First unified treatment identifying that semantic communication research has systematically neglected lower-tail reliability and proposing concrete design directions combining robustness with adaptive retransmission mechanisms.

Relevance to your work: Directly relevant to task-oriented communication research interests; the proposed joint design of robustness and adaptation under imperfect CSI connects to your work on AI-native air interfaces and learned PHY where reliability-performance tradeoffs are critical.

A Communication-Centric 6G-LLM Architecture for Scalable Tactical Autonomous Defense Vehicle Networks

Kiran Khurshid, Shumaila Javaid, Nasir Saeed · arXiv eess.SP · 2026-05-31

★★★★ Solid application-driven work demonstrating semantic communication benefits in a concrete multi-agent scenario, though simulation-only results and defense-specific scope limit immediate 6G standardization impact. This paper proposes a hierarchical architecture for tactical autonomous vehicle coordination that combines edge-hosted LLM reasoning with 6G semantic communication to reduce overhead and latency. Monte Carlo simulations across 5-30 vehicle fleets show 75% latency reduction (29ms vs 117ms) and 89% communication overhead reduction compared to 5G rule-based coordination. The approach uses semantic abstraction rather than conventional structured feature pipelines for multi-agent decision-making under contested network conditions.

Contribution: First demonstration of integrating edge-LLM semantic reasoning with 6G for tactical multi-agent coordination, showing dramatic reductions in both latency and communication overhead at scale.

Relevance to your work: Directly relevant as a concrete use case for semantic/task-oriented communication architectures, demonstrating how goal-level semantic abstraction can compress coordination traffic in beyond-5G networks with AI-native interfaces.

Hybrid Bit and Semantic Communications for UAV-Enabled Wireless Power Transfer Networks: A Decision-Assisted Deep Reinforcement Learning Approach

Jingfu Li, Jingjing Cui, Chong Huang, Jing Zhu, et al. · arXiv cs.IT · 2026-05-30

★★★ Competent application of semantic communications to UAV-WPT scenarios, but primarily combines existing concepts (semantic comms, WPT, DRL) without fundamental breakthroughs in semantic transmission methods or channel modeling.

This paper proposes a hybrid bit/semantic communication framework for UAV-enabled wireless power transfer networks, where UAVs harvest energy and adaptively switch between conventional and semantic transmission modes. The authors formulate a long-term optimization problem jointly optimizing UAV trajectory, base station selection, user association, transmission mode, and energy harvesting time to maximize semantic communication efficiency. They solve this using distributional soft actor-critic (DSAC) reinforcement learning with a decision assistant to improve convergence in dynamic SNR and task-varying conditions.

Contribution: Novel integration of adaptive semantic/bit communication mode selection with WPT in UAV networks, solved via DSAC with a decision assistant for improved DRL convergence.

Relevance to your work: Tangential—addresses semantic communication which intersects your interests, but focuses on network-layer resource allocation and energy harvesting rather than physical-layer semantic encoding, AI-native air interfaces, or advanced channel modeling techniques central to your work.

Gen AI based Channel Modeling

Scattering Environment Aware Joint Multi-BS Channel Estimation and Localization with Clock Asynchronism

Yani Chi · arXiv eess.SP · 2026-06-01

★★★★ Solid incremental work addressing a practical synchronization problem in distributed MIMO/ISAC systems using geometry-aided sparsity, though limited to simulation validation without experimental datasets.

The paper addresses clock asynchronism between base stations and users in multi-BS systems by exploiting shared scatterer information across BSs for joint channel estimation and localization. An AoA-based scatterer association algorithm provides coarse estimates of UE location and timing offsets, which are then refined via an EM framework that jointly estimates scatterer locations and channel coefficients. Simulations show improved accuracy over baseline methods in both channel estimation and localization.

Contribution: Novel multi-BS scatterer association using only AoA to bootstrap joint estimation of UE location, timing offsets, and scatterer positions despite clock asynchronism.

Relevance to your work: Directly relevant to ISAC research and massive MIMO channel estimation, particularly for distributed/cell-free architectures where clock synchronization and scatterer-based localization are key challenges.

AI-native air interface

AI-Based KPI Prediction Methods in Future 6G Networks: A Survey

Nilloofar Mehrnia, Gourav Prateek Sharma, Samie Mostafavi, Andreas Johnsson, et al. · arXiv eess.SY · 2026-06-01

★★★★ Solid survey work that organizes the KPI prediction landscape for 6G, but focuses on network management rather than fundamental physical-layer innovation.

This survey systematically reviews machine learning methods for predicting network Key Performance Indicators (KPIs) in 6G systems, covering capacity, latency, coverage, and reliability metrics. The authors introduce a multi-dimensional taxonomy classifying prediction approaches by KPI type, data source, protocol stack level, prediction horizon, model family, and objective. The work analyzes state-of-the-art methods from classical statistical models to deep learning and reinforcement learning, while discussing deployment challenges including data availability, scalability, privacy, and sustainability.

Contribution: First comprehensive taxonomy and systematic review of data-driven KPI prediction methods specifically for 6G networks, providing a structured framework spanning prediction objectives, model families, and deployment considerations.

Relevance to your work: Tangential – while relevant to AI-native 6G network automation, this addresses higher-layer network management and KPI forecasting rather than AI-native air interface design, physical-layer learning, or novel waveforms.